

FLEX-DBS manuscript: working notes

Working notes

Open items for the authors. Built separately from the manuscript (`make notes`); not part of the posted paper.

Data still needed

- **Electrode construction (Section 3.1).** Materials, wire diameter, insulation, exposed length or tip geometry, bilateral spacing, return electrode, connector and fabrication steps. Planned figure: electrode photograph and schematic.
- **Surgery and animals.** Strain, sex, age, housing, stereotaxic target and coordinates, anesthesia, IACUC protocol number, and the implantation date that anchors “weeks after implantation” in Section 3.
- **Battery measurements (Section 4.2, Table 3).** Measurement conditions are now stated (Joulescope JS220; 1 k Ω load and 180 μ s pulse width for the stimulating rows). Replace the runtime projections with a measured discharge to cutoff.
- **In-vivo validation (Sections 6.1–6.3).** Setup, wireless-versus-tethered comparison and behavioral outcome. Planned Figure 6 legend: (A) head-mounted device on the animal and arena; (B) example tracking; (C) behavioral metric for baseline, tethered DBS and FLEX-DBS wireless DBS, per animal and group.
- **Abstract and Conclusions.** Update the Results and Conclusions once in-vivo data exist; both currently describe design, 1 k Ω bench, impedance and battery results only.

Bench QC, 1 k Ω load (2026-10-01)

Three Joulescope JS220 recordings (500 kHz, $\sim$ 11 s each) at 10, 50 and 100 % amplitude, 180 μ s, 130 Hz set, 1 k Ω on both outputs, one output recorded (`data/bench/`, analyzed with `data/bench/bench_qc.py`). Results are in Section 4.1 and Figure 5 (`figures/scripts/fig_bench.py`). Firmware is not being changed now. Further bench work is deferred beyond the preprint (`future_revisions.md`).

Impedance measurement

- Data come from the platinum cohort 1 (M1, M2) and platinum cohort 2 (F1, F2) sheets. Excluded: M1 output 0 (near 0 V throughout, likely a short) and both electrodes of F1 (unstable; 47–118 k Ω and 23–106 k Ω between sessions, with within-session drift at 7 weeks). Analyzed: five electrodes from M1, M2 and F2. Confirm the wording of the exclusion criteria, and consider stating a quantitative criterion.
- The cohort 2 week-4 session was recorded at 4 weeks + 1 day and is plotted and analyzed as week 4.
- The StimJim bench check (2/10/20/68 k Ω) showed its voltage read-back 6–73 % below the oscilloscope, so the in-vivo values may be underestimates. Consider remeasuring with an impedance meter (for example, a NanoZ at 1 kHz) or with an oscilloscope on the StimJim output, and report both.
- The 450 μ s measurement pulse is longer than the DBS pulses; a measurement at the stimulation pulse width would better match the compliance calculation.

- The two-electrode configuration is justified with Koo and Weiland (2018); confirm that this is the framing the authors want.
- The other workbook sheets (gold-plated platinum, in vivo two months post, in vivo, bench testing) are not used.

Citations to add or verify

- Resolved 2026-10-01. Tether claims now cite Jiang et al. (2017) (movement and environment restriction) and Aulehner et al. (2022) (small measurable stress and sleep effects in rats), and the stress wording was softened to match. Slow-developing effects cite Kupsch et al. (2006) (delayed, progressive improvement with pallidal DBS for dystonia) and Aleksandrova et al. (2013) (premature responding took several sessions to emerge). The subthalamic-parameter sentence now cites Vachez and Creed (2020) instead of Creed (2018). Creed et al. (2012) and Creed et al. (2013) both used 100 μ A, 90 μ s, 130 Hz (Creed thesis, 2012, general methods); Aleksandrova et al. (2013) used 12.5 μ A, so the “several-hundred-microampere” claim in Section 2.3 was removed. Tucker et al. (2013) added for Howland output impedance and load dependence.
- Paulat et al. (2026) is still a Research Square preprint (posted 2026-04-27; checked 2026-10-01). Update if it is published.

Table 4

- Resolved 2026-10-01: de Haas et al. (2012) row filled from the full paper (Table 1 and Sections 2.1.6–2.1.8). Its output is a fixed-supply voltage source with a microcontroller-adjusted series resistance, entered as CC because the paper verifies current within 10 % into 15–45 k Ω . Kouzani parameters confirmed as 200 μ A, 90 μ s, 130 Hz. Burton uses the 25 kHz maximum, and the t-IPG uses the 5.3 mm thickness from the text of Paulat et al. (2026). Sizes were added to the mass column from each source; Alpaugh et al. (2019), Kouzani et al. (2013) (head-mount) and Burton et al. (2021) give no overall dimensions, so none is shown. Shen et al. (2026) does not state a channel count and was used unilaterally. The Pinnell Figshare release is described as design files and assembly instructions; whether it contains firmware source was not checked.
- Still open: supplementary files were not read for Burton et al. (2021) or Shen et al. (2026), and Paulat et al. (2026)’s programmable ranges and battery models come from its supplement.
- Inconsistencies within sources, shown in Table 4 with the alternate value in parentheses: Fleischer et al. (2020) amplitude and frequency (text versus table) and board size (9 $\times$ 9 mm table versus 10 $\times$ 10 mm abstract); Pinnell et al. (2018) mass (2.8 g table versus 2.7 g text). Not shown: Shen et al. (2026) gives $\geq$ 300 h in its table but more than 30 days in its discussion (table value used), and demonstrates settings outside its stated ranges.
- Kouzani et al. (2013) was designed for mice but tested in rats; Alpaugh et al. (2019)’s “wireless” means untethered, with no radio; its pulse width is read from its Figure 1d.

Firmware and design

- Firmware v0.5 delivers the 130 Hz setting as 142.5 Hz, and at 180 μ s the stimulating phase measures $\sim$ 60 μ s long and the recovery phase 22–46 μ s short, leaving a 26–35 % net charge per pulse (Section 4.1). If the fixes listed in Section 7 are made before posting, remeasure and update Section 4.1, Figure 5, Table 2 and the Limitations.

- The photograph in Figure 2A shows the “v8.1” board silkscreen; decide whether to mention the hardware revision in the text.
- Supplementary S1: consider adding the BOM (v8.1/BOM.csv) alongside the schematic.

End matter

- Confirm author order, middle initials and affiliations.
- Author contributions, competing interests, funding and acknowledgments are pending.
- Data availability: decide whether to deposit the impedance data and figure scripts in a public repository instead of “available on request”.
- Before journal submission (not needed for bioRxiv): Highlights, graphical abstract, CRediT statement, generative-AI declaration, line numbers and double spacing per the Journal of Neuroscience Methods guide.

References

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